

Enumeration

Its Active Connection to a Device.

Its the detailed information gathering.

Objectives

1. Explain Enumeration Concepts
2. Demonstration of Different Techniques for NetBIOS Enumeration.
3. Demonstrate Different Techniques for SNMP and LDAP Enumeration.
4. Use Different Techniques for NTP and NFS Enumeration.
5. Demonstrate Different Techniques for SMTP and DNS Enumeration.
6. Demonstrate Ipsec, VoIP, RPC, UNIX/Linux, and SMB Enumeration.

1. Enumeration Concepts:

It involves an attacker creating active connections with a target system and performing directed queries to gain more information about the targeted system

Attacker can use extracted info to identify points for a system attack and perform password attacks to gain unauthorized access to information system resources.

These techniques are conducted in an intranet environment.

Info Enumerated by Intruder:

- Network Resources
- Network Shares
- Routing Tables
- Audit and Service Settings
- SNMP and FQDN details
- Machine Names

- Users and Groups
- Applications and Banners

2. NetBIOS Enumeration

It came into 1990s just to make networking between systems.

Its present in Windows

A NetBIOS name is a unique 16 ASCII character string used to identify the network devices over TCP/IP; 15 characters are used for the device name, and the 16th is reserved for the service or name record type.

Attacker use NetBIOS enum to obtain the list of computers that belong to a domain, list of shares on the individual hosts in the network, policies and passwords...

NetBIOS Name List:

NAME	NetBIOS CODE	TYPE	INFO OBTAINED
<host name>	<00>	UNIQUE	Hostname
<domain>	<00>	GROUP	Domain Name
<host name>	<03>	UNIQUE	Messenger Service running for the Computer
<username>	<03>	UNIQUE	Messenger Service running for the logged-in user
<host name>	<20>	UNIQUE	Server Service running
<domain>	<1D>	GROUP	Master Browser name for the subnet
<domain>	<1B>	UNIQUE	Dom master browser name, identifies primary domain controller(PDC) for the domain

Command: nbstat -a <targIP> //for windows

nmap -sV -v --script nbstat.nse <targ_IP> // for linux

3. Techniques for SNMP and LDAP Enumeration

Attackers use SNMP default strings to extract info about a device.

Attackers enumeartes SNMP to extract information about network resources, such as hosts, routers, devices and shares, and network information, such as ARP tables, routing tables, and traffic.

SNMP=simple network manager protocol used to manage and monitor network devices.

SNMP-managed Devices:

.Routers

.Switches
.Firewalls
.Printers
.Servers
.Network Enabled IoTs

Why SNMP?

In real networks, admins needs **centralized visibility**.

Insted of logging into every device manually, SNMP lets tools collect data automatically and monitoring systems can alert on failure. **It uses 161 port number.**

Why Attackers' fav?

Because it uses UDP, generate minimal logs and often ignored by monitoring tools. Its trusted protocol so allowed by firewall internally and overlooked by admins. It can leak a lot of data as it act as a manager who knows more than others.

Tools:

1. **SnmpWalk:** `snmpwalk -v1 -c public <targIP>`
2. **Nmap:** `nmap -sU -p 161 --script=snmp-processes <targIP>`
3. **snmp-check**
4. **SoftPerfect Network Scanner**
5. **OpUtils**

Now lets move towards **LDAP:**

LDAP:

its is used in the AD.

Lightweight Directory Access Protocol is an internet protocol for accessing distributed services.

A client starts a LDAP session by sonnecting to a Directory System Agent(DSA) on TCP port 389 and then sends operation request to the DSA.

Attacker query the LDAP service to gather information such as valid usernames, addresses, and departmental details, which can be further used to perform attacks.

So we can also exploit LDAP to create a User Account for future attacks.

4. NTP and NFS Enumeration:

NTP=123=Network Time Protocol= you can pull out time server information such as list of connected hosts and clients' IP Addresses in a network, their system names and Oss.

NFS system is generally implemented on the computer networks, where the centralization of the data is required for critical resources. It enables attackers to identify the exported directories, list of clients connected to the Nfs server along with their IPs, and the shared data associated with the Ips

Tools:

1. **rpcinfo -p <targIP>**

5. SMTP and DNS Enumeration

SMTP=Simple Mail Transfer Protocol=25

Command: **nmap -p 25 --script=smtp-enum-users <targIP>**

DNS Enum:

What DNS leaks:

- Internal architecture
- Naming conventions
- Business logic

a DNS zone is

A DNS zone is a logical portion of the DNS namespace that a DNS server manages.

Concept	Meaning
DNS zone	Collection of DNS records
Zone transfer	Full zone replication
Risk	Information disclosure
DNS enumeration	Mapping infrastructure
Importance	Precision targeting

DNS zone transfer vs brute-force subdomain discovery

Method	Accuracy	Noise
Zone transfer	100%	Very low

Method	Accuracy	Noise
Brute-force guessing	Partial	High

That is why zone transfer exposure is dangerous.

Why attackers LOVE DNS zone transfers

Because it gives:

- Complete reconnaissance
- Zero guessing
- No brute forcing
- No scanning noise

One mistake = full visibility.

Tools:

1. Amass:

- **Passive Enum:** `amass enum -passive -d <targDomain/IP> -src`
- **Active Enum:** its uses brute-forcing command is= `amass enum -active -d <targDom> -brute -w <wordlistPath>`

2. Nmap

6. IPSec, VoIP, RPC, UNIX/Linux and SMB Enumeration

VPNs are based on IPSec at UDP port 500.

Attacker can probe further using a tool, such as **ike-scan**, to enumerate sensitive information, including encryption and hashing algorithms, auth type, key distribution algo etc

Command: `ike-scan -M <targIP>`

VoIP: uses Session Initiation Protocol (SIP) to enable voice and video calls over an IP networking.

SIP generally uses UDP/TCP ports 2000, 2001, 5060, and 5061.

Tools:

1. SVMAP: `svmap <targIP>`

SMB: tools: `enum4linux <targIP>`